

Abhinav Mahajan

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EDUCATION

Carnegie Mellon University

Master of Science in Computer Vision

Pittsburgh, PA

December 2026

International Institute of Information Technology

(Dual Degree) Master and Bachelor of Technology in Electronics and Communication Engineering

Bangalore, India

July 2025

Dual specialization: Artificial Intelligence/Machine Learning and Network Communication

GPA: 3.63/4.00; Dean's Merit List; Teaching Assistant for courses Machine Learning (AI 511), Visual Recognition (AI 825)

PROFESSIONAL EXPERIENCE

Amazon

Applied Scientist Intern

Bangalore, India

January 2025 - June 2025

- Developed a novel exposure mitigation strategy and long sequential input context reduction technique in LLM's for extracting Customer-Interest Category propensity scores for bi-directional ranking for personalized AD targeting.
- Achieved 3 times greater recall in next-purchase interest category prediction with minimal precision drop; Currently in pre-production and ready for submission in ACM SIGMOD/PODS 2026 (Applied Track).

Adobe Research

Research Intern

Bangalore, India

May 2024 - July 2024

- Devised an automated pipeline inspired by design processes employed by professionals to enhance visual appeal of Graphic designs (Posters/Flyers) introducing seminal techniques in harmonizing image assets while retaining identity.
- Approved for patent (USPTO filing expected Sept 2026); Under Review at a prestigious Computer Vision conference.

Adobe Research

Research Intern

Bangalore, India

May 2023 - August 2023

- Engineered a self-supervised graphic design evaluation system to overcome annotation bias and crafted a novel refinement algorithm to optimize layout aesthetics via the learned scorer; System deployed in Adobe Express.
- Outperformed GPT-4o by 12% on design evaluation tasks despite being 600 times smaller; paper accepted in WACV 2025 (Round 1, 13% acceptance rate).

PUBLICATIONS

[1] **Abhinav Mahajan***, Abhikhya Tripathi*, KJ Joseph et al. *Make-It-Pretty: Towards Generating Designs from its Components*, **Under Review**

[2] Sahil Goyal*, **Abhinav Mahajan***, KJ Joseph et al. *Design-o-meter: Towards Evaluating and Refining Graphic Designs*, **WACV 2025**

[3] Ali Abedi*, Tracey JF Colella*, Mark Bayley, Urvashy Gopaul, **Abhinav Mahajan**, Tarun Reddy et al. *AVA: AI-driven Virtual Rehabilitation Assistant*, **WCISVR 2023**

PROJECTS

Hierarchical Sentiment and Activity Analysis of Users

August 2024 - October 2024

- Awarded the MITACS Fellowship and interned at Tech University of Ontario, Canada. Curated and annotated a dataset using screenshot parsing and vision-language models to infer user activity from screen captures.

Pothole Detection and Severity Estimation

December 2022 - January 2023

- Won SDAIA Smartathon Hackathon among 1,000+ submissions by training a Mask R-CNN model to segment potholes from video data with 82% accuracy and 3D reconstruction of segmented potholes to assess severity.

Stroke Rehabilitation Agent

May 2022 - December 2022

- Collaborated with University of Toronto to build a video-to-animation script enabling clinician-prescribed exercises to be rendered to avatars; Website deployed allowing stroke patients to attend rehabilitation sessions virtually.
- Developed a novel cyclic interpolation algorithm to achieve jitter-free animations; Work accepted to WCISVR 2023.

SKILLS

Programming Languages: Advanced: Python; Basic: C++, C, Java

Libraries: Advanced: PyTorch, NumPy, Pandas, Huggingface Transformers, trl; Basic: Tensorflow